Advanced Causal Inference

Lecture 02: Neyman-Rubin Causal Model

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This lecture note is partially based on the lecture notes/textbook of Hyunseung Kang, Ying Zhou, Ding (2024), Hernán and Robins (2020)

Outline

1. Neyman-Rubin Causal Model
2. History of Causal Frameworks
3. Fundamental Problem of Causal Inference
4. Causal Estimands
5. Manipulability

Neyman-Rubin Causal Model

- The model was developed by Neyman (1923), republished in Neyman et al. (1990), and Rubin (1974)
- See Holland (1986) for more background and history

Treatment & Potential outcomes & Individual Causal Effect

- Treatment A: an intervention or exposure assigned to each unit (person, place, or thing which may receive treatment)
  - $A = 1$: a unit receives the treatment
  - $A = 0$: a unit receives the control
- Treatment should be manipulable; no causation without manipulation (Holland, 1986) (which we will discuss later)
- Potential outcome $Y(a)$: the outcome of a unit had, possibly contrary to fact, it received $A = a$
- Individual causal effect: $\tau_i = Y_i(1) - Y_i(0)$

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- Individual causal effect: $\tau_i = Y_i(1) - Y_i(0)$
- $A =$ Smoking, $Y =$ Lung cancer
- $Y_i(1) = \mathbb{1}$(Person i has lung cancer if she smoked)
  $Y_i(0) = \mathbb{1}$(Person i has lung cancer if she did not smoked)
- $\tau_i = 1$ smoking causes lung cancer for person i
  $\tau_i = 0$ smoking does not cause lung cancer for person i
  $\tau_i = -1$ smoking prevents lung cancer for person i

Caveats

- Causal effect of a treatment can only be defined in comparison to another treatment, where one treatment may be inactive for unit i
- It is easier to think about $Y_i(1)$ and $Y_i(0)$ using a parallel universe analogy.
  - Suppose, in one universe, unit i is receiving treatment and in another exactly identical/parallel universe, unit i is receiving control.
    The potential outcomes $Y_i(1)$ and $Y_i(0)$ reflect the outcomes from these two universes, but everything else remains the same about unit i.
    Because the only difference between the two parallel universes are their treatment assignment status, then any difference in the outcomes from the two parallel universes can only be attributed to the differences in the treatment assignment.
- The interpretation is similar to "after accounting for all other variables" interpretation of regression coefficients that we learned in linear regression courses

Caveats

- By definition, τ_i is the **effects of causes** (e.g., effect of smoking vs. not smoking) rather than the **causes of effects** (e.g., why does he have lung cancer?). The causes of effects are hard to define because of the problem of infinite regress.
  - Rubin's example: He got lung cancer because he smoked cigarettes. The real reason he smoked is because his parents smoked, and they smoked because they hated each other and they hated each other because $\cdots$
- The potential outcomes model provides a clear definition of the effects of causes, and it provides actionable information on how to live our lives (i.e., if you know that smoking results in lung cancer, you may choose not to smoke).
- Cause-effect relationships naturally have a temporal ordering: treatment variable always precedes the *observed* outcome variable. In particular, You cannot have a result (i.e. outcome) before a cause (i.e. treatment).
- You also cannot have causal simultaneity where the outcome and treatment variable simultaneously change each others values at the exact same time. This makes it impossible to determine whether the outcome is causing the treatment variable or vice versa.
- The causal effect τ_i is defined as a linear difference. But, causal effects can also be defined based on other scales, e.g., log scale.

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Hume's Definiton on Causality



We may define cause to be an object followed by another, and where all the objects similar to the first are followed by objects similar to the second. Or in other words where, **if the first object had not been, the second never had existed**

David Hume (1748)

James Lind and His Pioneering Medical Trial (1747)



- One of the first recognized medical trials
- Aim to understand the relationship between eating oranges and lemons and recovering from scurvy (which occurs when vitamin C is deficient)
- Twelve men showing advanced symptoms of scurvy were divided into six pairs, matched "as closely as possible," to receive different treatments
  - Oranges & Lemons: Significant recovery in just a week
  - Cider: Slight improvement; still weak after two weeks
  - Elixir of vitriol: No notable improvement
  - Vinegar: No notable improvement
  - Sea water: No notable improvement
  - Medicinal paste & Cream of tartar: No notable improvement

Jerzy Neyman (1923)



- **Design of field experiments**:
  - Field divided into N equal plots
  - Each plot has true yields: $U_1, U_2, \ldots, U_M$ from M crop varieties
- **Neyman's Urn Analogy**:
  - M urns, one for each variety
  - In the ith urn, N balls, each labeled with potential yield of the ith variety on a respective plot: U_{ij} where $i = 1, \ldots, M$ and $j = 1, \ldots, N$
  - Under the current potential outcome framework, we would write $Y_j(i)$

Donald Rubin's Contribution



- Neyman first proposes potential outcomes in his master thesis!
- But he refuses to acknowledge that it can be used in observational studies
- Rubin extends the potential outcome framework to observational study settings and proposes the SUTVA (which we will discuss later)

Other frameworks for defining the causal effect

- Three commonly used frameworks for causal inference
- The potential outcome framework (Commonly used in statistics)
- Structural Equation models (Commonly used in econometrics)
- Graphical models (a.k.a. Bayesian networks) (Commonly used in computer science)
- (in my view) They are mostly equivalent, though each has distinct established goals, tools, and areas of application.
- However, some argue that one is essential for causal reasoning;
  see this blog and its comments for a heated discussion.

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Relationship Between Potential and Observed Outcomes

- Each individual has two potential outcomes $Y(1)$ and $Y(0)$ (assuming binary action A)
- In contrast, we typically only observe one outcome Y for each unit
- An elementary assumption is needed to connect these two outcomes

Stable Unit Treatment Value Assumption (SUTVA)

Assumption (SUTVA (Rubin, 1980))

- *No interference between units (Cox, 1958)*
- *No different versions of a treatment; also known as (causal) consistency (Cole and Frangakis, 2009).*

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   - Interference: the potential outcome $Y_i(a)$ for an individual i depends on what treatment other people receive
   - Examples: vaccination, advertising, infectious diseases, social networks, agricultural experiments
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2. Consistency: No different versions of a treatment
   - Sometimes, a treatment A doesn't have a clear meaning, as it has many versions
   - Example: Different surgeons performing the same procedure
   - Rule out the case that potential outcomes depend on the specific version of the treatment received

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- Under SUTVA, each unit has only two potential outcomes: $Y_i(1)$ and $Y_i(0)$.
- $Y_i = A_i Y_i(1) + (1 - A_i) Y_i(0) = Y_i(A_i) \quad \Leftrightarrow \quad$ if $A_i = 1$ then $Y_i = Y_i(1)$; if $A_i = 0$ then $Y_i = Y_i(0)$

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- SUTVA connects the intervention we see (A) with the causal intervention of interest $(A = a)$

The Fundamental Problem of Causal Inference

- Under SUTVA, only one potential outcome is observed:

	$Y(1)$	$Y(0)$	$Y(1) - Y(0)$	A	Y
Zeus	1	0	1	1	1
Hera	0	0	0	1	0
Hebe	0	1	-1	0	1
Apollo	0	0	0	0	0
Ares	1	1	0	1	1

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- Because we never observe both potential outcomes, they are sometimes referred to as "counterfactuals"
  - However, under SUTVA, one of the potential outcomes is actually observed (the factual), which makes the term "observed counterfactuals" (?!)
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  - To avoid confusion, it is better to simply use the term "potential outcomes"
- We cannot identify the individual causal effect for any individual! (unless strong assumptions are made)

Before-After Study

- Suppose you drink a cup of coffee at 8:00. One minute later, at 8:01, you check how awake you feel. You notice that you do feel more awake. Thus, you conclude that drinking coffee caused you to feel more awake.
- $Y_i(0)$: How awake you would feel at 8:01 if you did not drink coffee at 8:00
  $Y_i(1)$: How awake you would feel at 8:01 if you did drink coffee at 8:00
- $\tau_i = Y_i(1) - Y_i(0)$

Before-After Study

- Temporal Stability Assumption:
  - Your level of wakefulness without coffee at 8:01 would be the same if we measured it at any similar time; the time you measure does not change what happens naturally.
  - So, if you had NOT drunk coffee at 8:00, your alertness level at 8:01 would be the same as if you checked your alertness at 7:59, assuming nothing else changed.

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- Causal Transience Assumption:
  - Your alertness when you have not had coffee does not change how awake you would be after drinking coffee, and vice versa.
  - For example, if you check your alertness at 7:59 before drinking coffee, this check does not affect how awake you will feel after drinking coffee at 8:00.

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- We can measure both $Y_i(0)$ and $Y_i(1)$ if

(i) when you measure them does not affect their values (temporal stability) and
(ii) measuring one value does not affect the other value (causal transience)

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- Disease progression:
  - Many illnesses improve naturally over time, so treatments may appear effective even when they have no real impact, as symptoms resolve on their own
  - For example, you get the flu on day 1, drink soda on day 2, and recover from the flu on day 3
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  - The temporal stability assumption is violated
- Hawthorne Effect:
  - Phenomenon where individuals modify their behavior when they are aware of being observed
  - For example, a CEO observes a worker's productivity on day 1, installs a new production machine on day 2, and notices an improvement in productivity on day 3
  - The causal transience assumption is violated

Lab-Controlled Experiments and Unit Homogeneity

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- Under unit homogeneity, we can apply treatment to unit i and control to unit j. Then, the causal effect of treatment compared to control on unit i (and also unit j) is taken to be $Y_i(1) - Y_j(0)$.
- Unit homogeneity is frequently used to make causal inferences in scientific laboratories.

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- Unit homogeneity is frequently used to make causal inferences in scientific laboratories.
- Example: Knockout experiments on mice
  - Knockout mice are used to study how modifying a specific gene (treatment = gene knocked out; control = normal gene) affects the development of diseases.
  - For example, p53 knockout mice lack the p53 tumor suppressor gene.
  - Scientists have observed that these mice have a significantly higher likelihood of developing tumors compared to normal mice. This analysis relies on the unit homogeneity assumption: mice with and without the gene respond consistently to the knockout.
  - Mario Capecchi, Martin Evans, and Oliver Smithies, who developed the knockout mouse technology, were awarded the Nobel Prize in Physiology or Medicine in 2007.

Lab-Controlled Experiments and Unit Homogeneity

- When does unit homogeneity not hold?
- If study units are humans. Humans are not all the same (even twins) and some have differential responses to treatment/control.
- If study units cannot be factory-replicated, either in a bio or engineering lab.

Summary

- Under SUTVA, we observe only one of the potential outcomes for each unit
- Consequently, the individual causal effect is not identifiable
- While exceptions exist, the required assumptions are often unrealistic
- Need to target an estimand that can be identified

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Estimand: Asking the Right Question

A man decides to wallpaper his bedroom but isn't sure how many rolls he'll need. His next-door neighbor recently did the same job, and their rooms are identical in size.

"How many rolls of wallpaper did you buy for your bedroom?" he asks.
"Ten," the neighbor replies.

The man buys ten rolls, does the job, and it looks perfect—except he has four rolls left over.

"Hey, neighbor," he says. "I bought ten rolls like you said, but I've got four left over."

"That's funny," the neighbor says.

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"Hey, neighbor," he says. "I bought ten rolls like you said, but I've got four left over."

"That's funny," the neighbor says. **"So did I."**

Average Treatment Effect

- **Average Treatment Effect (ATE):**

$$\tau_{\mathsf{ATE}} = \mathrm{E}\{Y(1) - Y(0)\} = \mathrm{E}\{Y(1)\} - \mathrm{E}\{Y(0)\}$$

- We do not need the *joint* distribution of $\{Y(1), Y(0)\}$;
  only need the *marginal* expectations $\mathrm{E}\{Y(1)\}$ and $\mathrm{E}\{Y(0)\}$

- Analogy: survey sampling
  Suppose we want to describe the distribution of student heights at UIUC. We could:

(i) Measure *everyone's* height and construct the full histogram; or

(ii) Draw N random samples and estimate the mean height as a "good enough" summary

- Learning the *individual treatment effect* is like option (i): highly labor-intensive, requiring full information for each individual (analogous to estimating τ_i for all i), and relying on strong, unrealistic assumptions

- Learning the *ATE* is like option (ii): less detailed than (i), but supported by well-developed statistical methods that quantify estimation uncertainty

- In fact, option (ii) is the basis for randomized clinical trials in medicine for drug/therapy approvals

Other Popular Estimands

- Effects defined by linear contrasts:
- Average treatment effect (ATE): $\tau_{\mathsf{ATE}} = \mathrm{E}\{Y(1) - Y(0)\}$
- Average treatment effect on the treated (ATT): $\tau_{\mathsf{ATT}} = \mathrm{E}\{Y(1) - Y(0) \,|\, A = 1\}$
- Average treatment effect on the control (ATC): $\tau_{\mathsf{ATC}} = \mathrm{E}\{Y(1) - Y(0) \,|\, A = 0\}$

- Some nonlinear effects:
- Quantile treatment effect: $\tau_{\mathsf{QTE}} = q_\alpha(Y(1)) - q_\alpha(Y(0))$ where $q_\alpha(V)$ is the αth quantile of a random variable V
- Relative risk: $\tau_{\mathsf{RR}} = \dfrac{\mathrm{E}\{Y(1)\}}{\mathrm{E}\{Y(0)\}}$
- When Y is binary, odds ratio: $\tau_{\mathsf{OR}} = \dfrac{\Pr\{Y(1) = 1\}}{\Pr\{Y(1) = 0\}} \dfrac{\Pr\{Y(0) = 0\}}{\Pr\{Y(0) = 1\}}$

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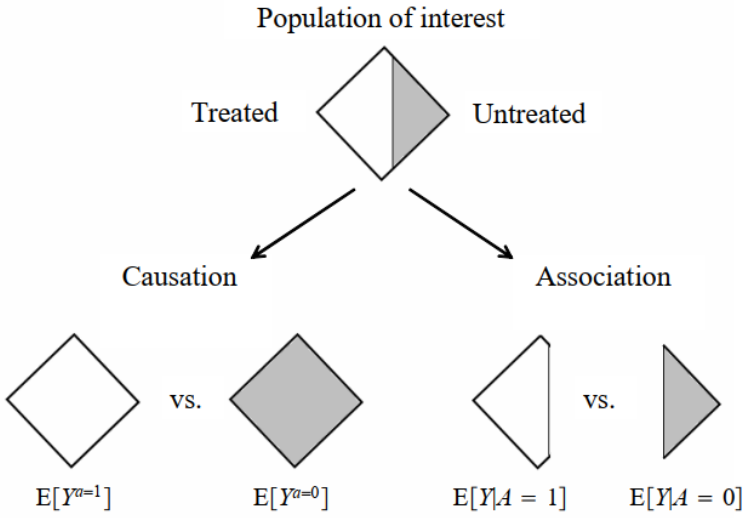
- Average treatment effect (ATE): $\tau_{\mathsf{ATE}} = \mathrm{E}\{Y(1) - Y(0)\} = 0$
- Average treatment effect on the treated (ATT): $\tau_{\mathsf{ATT}} = \mathrm{E}\{Y(1) - Y(0) \,|\, A = 1\} = 1/3$
- Average treatment effect on the control (ATC): $\tau_{\mathsf{ATC}} = \mathrm{E}\{Y(1) - Y(0) \,|\, A = 0\} = -1/2$
- Quantile treatment effect: $\tau_{\mathsf{QTE}} = q_\alpha(Y(1)) - q_\alpha(Y(0)) = 0$
- Relative risk: $\tau_{\mathsf{RR}} = \dfrac{\mathrm{E}\{Y(1)\}}{\mathrm{E}\{Y(0)\}} = 1$
- Odds ratio: $\tau_{\mathsf{OR}} = \dfrac{\Pr\{Y(1) = 1\}}{\Pr\{Y(1) = 0\}} \dfrac{\Pr\{Y(0) = 0\}}{\Pr\{Y(0) = 1\}} = 1$

Causation v. Association

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- Average treatment effect (ATE): $\tau_{\text{ATE}} = \text{E}\{Y(1) - Y(0)\} = 0$
- Difference-in-means: $\text{E}(Y|A = 1) - \text{E}(Y|A = 0) = 2/3 - 1/2 = 1/6$
- Association $\not\Rightarrow$ Causation

Causation v. Association



Causation v. Association

- In fact, $\mathrm{E}\{Y(a)\} \neq \mathrm{E}(Y \mid A = a)$, from both a mathematical and interpretive perspective
  - $\mathrm{E}\{Y(1)\}$ and $\mathrm{E}\{Y(0)\}$ represent averages of potential outcomes for the same population under treatment and control, respectively. To put it differently, in both expectations, we are taking average values of Y among the same set of individuals.
  - Then, the ATE represents the average among the same population of units
  - If the ATE is not zero, it can only be attributed to the difference in the treatment variable since everything except for their treatment assignment differs between the two expectations
  - $\mathrm{E}(Y|A = 1)$ and $\mathrm{E}(Y|A = 0)$ represent averages of outcomes among units who received treatment and control, respectively
  - While they may sound like they are averaging over the same population, each conditional expectation may represent averages of outcomes among two (potentially different) subpopulation of units
  - Thus, the difference in means of Ys among treated and control units $\mathrm{E}(Y|A = 1) - \mathrm{E}(Y|A = 0)$ are differences in averages of Ys from two potentially different subpopulations
  - If this difference is non-zero, it may not only be due to different treatment assignments, but also differences in the two subpopulations represented by the conditional expectations

Causation v. Association

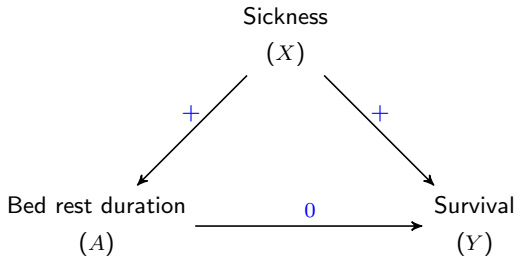
- Causation: compare two hypothetical worlds
  - What would be the risk if everybody had been treated?
  - What would be the risk if everybody had been untreated?

- Association: concerned with questions in the actual world
  - What is the risk in the treated?
  - What is the risk in the untreated?

- We can only see the association, but we want to infer causation

Source of Association

- There are two reasons that may lead to association between A and Y
  - **Causal effect**: the quantity we aim to estimate
  - **Confounding**: systematic differences between treated and untreated units



Kidney Stone Treatment Example

- Treatment (A):
  - 1: Open surgical procedure
  - 0: Small puncture procedure
- Outcome (Y): Success (1) or failure (0)

Aggregated data:

	$Y = 1$	$Y = 0$
$A = 1$	273	77
$A = 0$	289	61

Kidney Stone Treatment Example

- Treatment (A):
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	$Y = 1$	$Y = 0$
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- Stone size (L): Larger stones (1) or smaller stones (0)

Smaller stones:

	$Y = 1$	$Y = 0$
$A = 1$	81	6
$A = 0$	234	36

Larger stones:

	$Y = 1$	$Y = 0$
$A = 1$	192	71
$A = 0$	55	25

Observations

Aggregated data:

	$Y = 1$	$Y = 0$
$A = 1$	273	77
$A = 0$	289	61

Smaller Stones:

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- Aggregated data suggests Treatment 0 is more effective:

$$\widehat{\tau}_{RR} = 78\% - 83\% = -5\%$$

- Subgroup data suggests Treatment 1 is more effective:
  - Smaller stones: $\widehat{\tau}_{RR} = 93\% - 87\% = 6\%$
  - Larger stones: $\widehat{\tau}_{RR} = 73\% - 69\% = 4\%$

Yule-Simpson Paradox

- The association between two variables (A and Y) reverses upon conditioning on a third variable (X)
- Marginal association: negative
- Conditional associations: positive in both subgroups

Explanation of the Paradox

Smaller Stones:

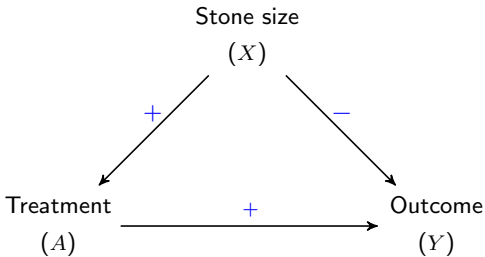
	$Y = 1$	$Y = 0$
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Larger Stones:

	$Y = 1$	$Y = 0$
$A = 1$	192	71
$A = 0$	55	25

- **Confounding Variable:** Stone size
- Patients with large/small stones are more likely to receive Treatment 1?
- Patients with large/small stones are more likely to have success?
- Illustrated by the DAG (next slide).

DAG of the Kidney Stone Example



- The signs indicate the associations of two variables conditioning on other variables pointing to the downstream variable

Key Takeaways

- The Yule-Simpson Paradox demonstrates the importance of stratifying data (i.e., accounting for confounding)
- Aggregated data can mislead causal interpretations due to confounding
- Always consider potential lurking variables when analyzing data

Yule-Simpson Paradox cannot happen to causal effects

- In the observed data:

$$\begin{array}{ccccccccc}
\Pr(Y=1|A=1) & = & \Pr(Y=1|A=1,X=1) & \times \Pr(X=1|A=1) & + & \Pr(Y=1|A=1,X=0) & \times \Pr(X=0|A=1) \\
\wedge & & \vee & \vee & & \vee & \wedge \\
\Pr(Y=1|A=0) & = & \Pr(Y=1|A=0,X=1) & \times \Pr(X=1|A=0) & + & \Pr(Y=1|A=0,X=0) & \times \Pr(X=0|A=0)
\end{array}$$

- In the counterfactual data:

$$\begin{array}{ccccc}
\Pr\{Y(1)=1\} & = \Pr\{Y(1)=1|X=1\}\Pr(X=1) & + \Pr\{Y(1)=1|X=0\}\Pr(X=0) \\
\vee & \vee & \vee \\
\Pr\{Y(0)=1\} & = \Pr\{Y(0)=1|X=1\}\Pr(X=1) & + \Pr\{Y(0)=1|X=0\}\Pr(X=0)
\end{array}$$

Outline

1 Neyman-Rubin Causal Model

2 History of Causal Frameworks

3 Fundamental Problem of Causal Inference

4 Causal Estimands

5 Manipulability

Manipulability

- Revisiting the earlier discussion:

  Treatment should be manipulable; no causation without manipulation (Holland, 1986) (which we will discuss later)

- Examples:
  - Does anti-retroviral therapy affect CD4 count in HIV-positive individuals?
  - Does caloric intake affect CD4 count?
  - Does obesity affect CD4 count?
  - Does race affect CD4 count?

Example of An Ill-defined Treatment: Race

- Racial discrimination is an important issue in the labor market
- However, *race* as a treatment is not well-defined:
  - It is not easy to imagine an experiment that changes the race of an experimental unit
- Example: Bertrand and Mullainathan (2004) (with more than 8000 citations)
  - The study investigates racial discrimination in hiring practices
  - They randomly assign CVs with **African-American-sounding names** (e.g., Lakisha Washington or Jamal Jones) or **White-sounding names** (e.g., Emily Walsh or Greg Baker
  - The treatment: Binary indicator for a Black-sounding or White-sounding name
  - The outcome: Binary indicator of receiving a callback

Results from Bertrand and Mullainathan (2004)

- Observed Data (2x2 Table):

	Callback	No Callback	Total
Black	157	2278	2435
White	235	2200	2435

- Comparison of Probabilities:

$$\frac{157}{2278+157} - \frac{235}{2200+235} = 6.45\% - 9.65\% = -3.20\% \quad (< 0)$$

- Statistical Test:
  - Fisher's exact test p-value: much smaller than 0.001

Key Takeaways

- In this experiment, the treatment is *perceived race*, which can be manipulated (e.g., by changing names on resumes)
- Critics may argue that the causal effect of perceived race may differ from the causal effect of actual race.
- This highlights the difficulty in defining treatments for abstract or non-manipulable concepts like race.

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